Week 3: Linux Logs, Sysmon, auditd and Snort Setup

Theory Notes, Developing Basic Understanding in SOC

Question 8. Create a comparison table of Windows Event Log vs Sysmon vs Linux auditd in terms of what they capture, log format, and use cases for a SOC analyst.

The table below compares the three sources side by side.

Point	Windows Event Log	Sysmon	Linux auditd
Platform	Windows	Windows, added as a tool	Linux
Available by default	Yes	No, it must be installed	Often needs to be installed, depending on the distro
What it captures	Logons, account changes, services, policy changes, system and application events	Process creation, network connections, file creation, registry changes, DNS queries, DLL loads, process access	System calls, command execution, file access and changes, logins, use of privileges
Process command line	Only through Event 4688 with an extra policy	Yes, in Event ID 1 along with hashes and parent process	Yes, through EXECVE and SYSCALL records
Log location	Event Viewer, with .evtx files in C:\Windows\System 32\winevt\Logs	Sysmon Operational log in Event Viewer	/var/log/audit/audit.log
Log format	Binary EVTX with XML data	EVTX with XML data	Plain text key=value records
Configuration	Audit policy and group policy	XML config file such as SwiftOnSecurity	Audit rules in /etc/audit/rules.d/
Main strength	Good for logins and account tracking, and always available	Deep endpoint detail, good for threat hunting	Detailed tracking of user actions on Linux
Main limit	Weak detail on processes and network	Needs tuning and can be noisy	Rules need care and logs can grow fast
SOC use cases	Failed logons (4625), successful logons (4624), new users (4720)	Spotting encoded PowerShell, command and control connections, dropped files	Tracking sudo commands, changes to sensitive files, user logins

In short

- I use Windows Event Logs to see who logged in and which accounts were changed.
- I use Sysmon to see what ran on a Windows machine and what it connected to.
- I use auditd to see what users did on Linux machines.
- When all three are used together, a SOC gets much better coverage than from any one of them alone.